

Olist E-Commerce

SQL Business Analysis

Ten business questions answered across revenue, geography, products, customers, payments, delivery, reviews, sellers, retention, and basket behavior - with explicit controls for fact grain and join multiplication.

DELIVERED REVENUE

R\$15.42M

item price plus freight

ORDERS ANALYZED

96.5K

delivered marketplace orders

SQL QUESTIONS

10

validated result extracts

CASE-STUDY PROMISE

- Keep item revenue, payments, reviews, and delivery events at their correct grains.
- Use CTEs and window functions to make transformations reviewable.
- Validate nine source tables before interpreting results.
- Translate result tables into decisions, not just query output.
- Provide PostgreSQL scripts and a server-free local execution path.

Model and Grain Controls

The analytical model is normalized; the SQL chooses the correct fact path for each question



WHY GRAIN MATTERS

- Joining payments directly to items can multiply both revenue facts.
- The source contains 547 orders with multiple review rows.
- Seller delivery rates must be order-weighted, not item-weighted.
- Repeat behavior requires customer_unique_id, not order-level customer_id.
- Geolocation repeats ZIP prefixes and is not joined without aggregation.

VALIDATION RESULT

1,550,922 rows loaded across nine tables. Every expected row count matched and all six tested orphan counts were zero.

Revenue Concentration

A large market with a concentrated geographic core

DELIVERED REVENUE

R\$15.42M

item revenue plus freight

SAO PAULO SHARE

37.4%

R\$5.77M delivered revenue

TOP-FIVE STATES

73.2%

share of delivered revenue

TOP AOV

R\$242.31

highest state-level AOV

TOP STATES BY DELIVERED REVENUE



BUSINESS READ

- Sao Paulo alone produces more than one-third of delivered revenue.
- The top five states form the natural first market for retention and service tests.
- Revenue concentration raises the impact of operational issues in the Southeast.
- Average order value varies by state, so order count and value should be managed separately.

Product Mix and Revenue Timing

Category breadth with a visible year-end promotional peak

TOP CATEGORIES BY PRODUCT REVENUE



The top five categories contribute 39.8% of product revenue. Health and beauty leads total revenue, while watches and gifts achieves high value with fewer units and a R\$199 average item price.

STRONGEST MONTHS - TOP TEN CATEGORIES

Month	Revenue
2017-11	R\$642.9K
2018-05	R\$613.9K
2018-03	R\$605.4K
2018-04	R\$581.0K
2018-01	R\$574.4K

INTERPRETATION

- November 2017 is the strongest observed month at R\$642.9K.
- March through May 2018 forms a sustained high-revenue period.
- Boundary months are partial, so the analysis avoids treating them as comparable full periods.
- Category mix supports differentiated pricing, inventory, and campaign strategies.

Delivery Reliability Drives Experience

Late orders show a severe shift toward low review scores

LATE-ORDER RATING

2.57

average review score

ON-TIME RATING

4.29

average review score

LATE ONE-STAR SHARE

46.1%

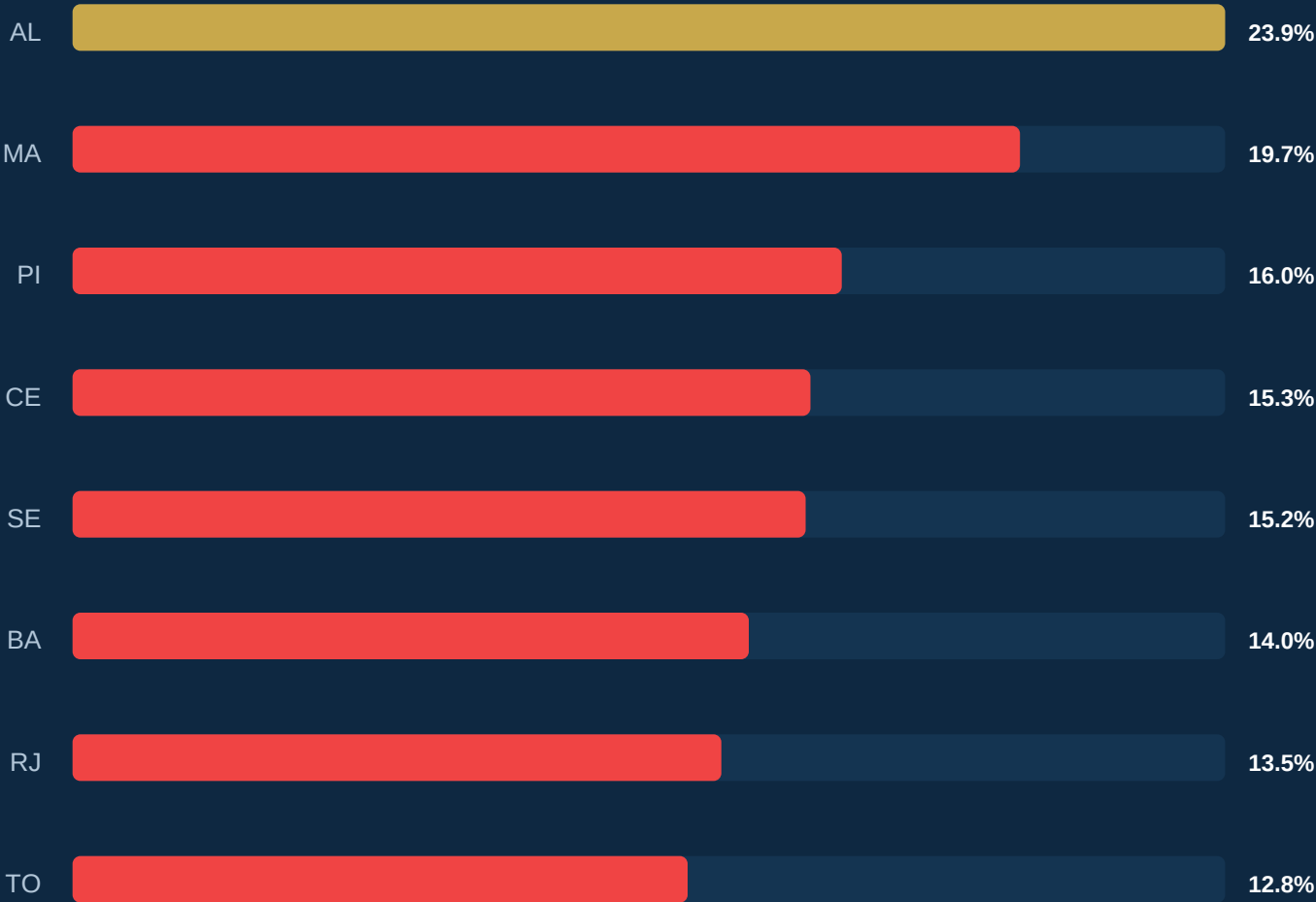
versus 6.6% on time

WORST STATE LATE RATE

23.9%

Alagoas; 397 orders

HIGHEST LATE-DELIVERY RATES



DECISION

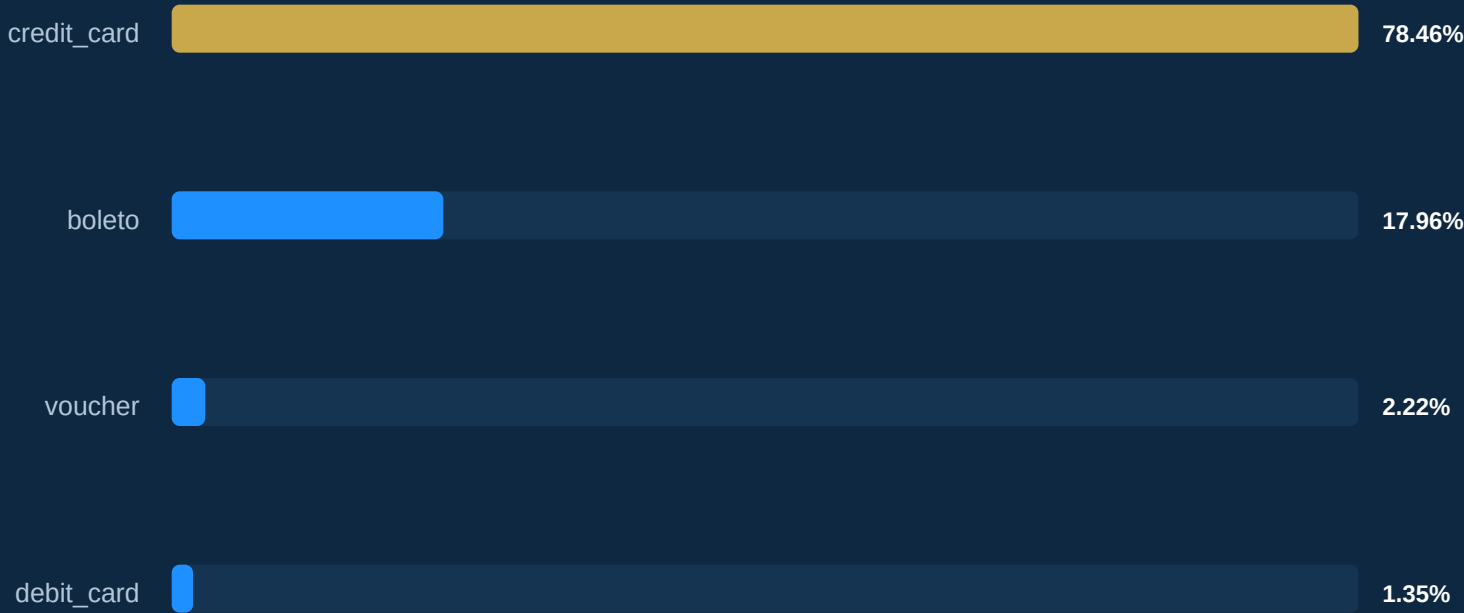
Prioritize carrier, route, and promise-date diagnostics in AL, MA, PI, CE, and SE. Use PR, MG, and SP as higher-scale operating benchmarks, where observed late rates remain near 5-6%.

- The relationship is associative, not a causal experiment.
- The magnitude still makes delivery reliability a high-confidence experience lever.
- Review analysis first collapses multiple review rows to the order grain.
- Track both late rate and total delivery days; they answer different operational questions.

Commercial Levers

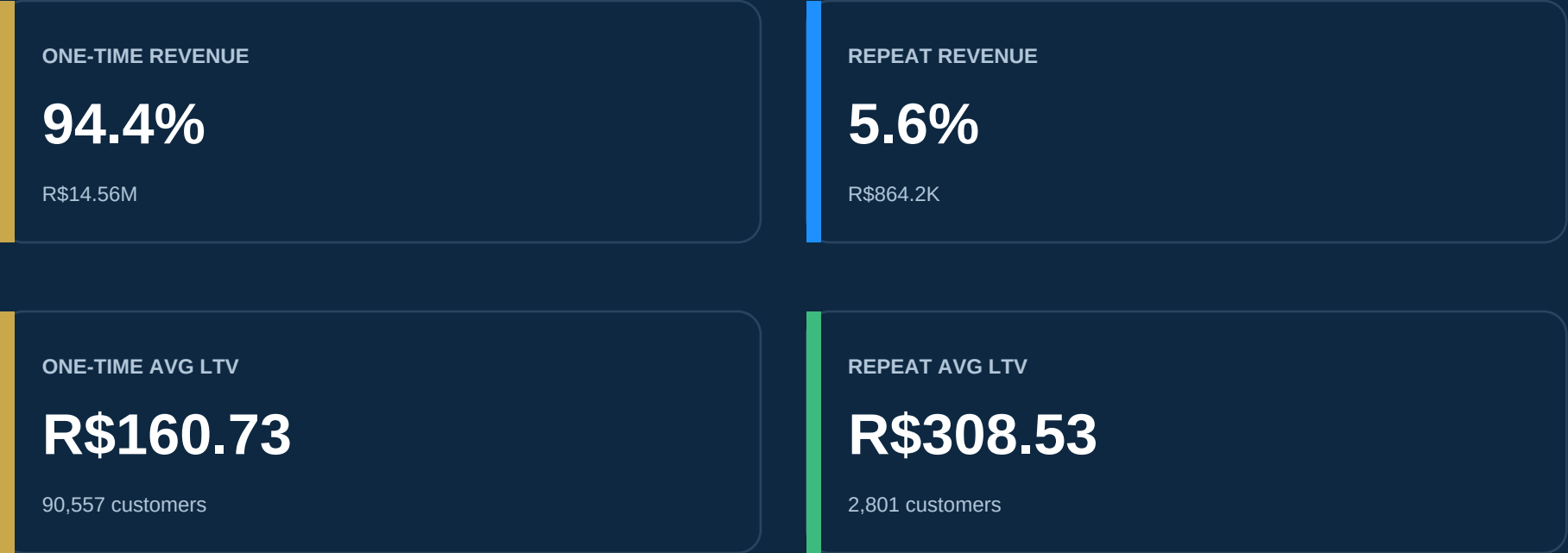
Payment dependence and a large second-purchase opportunity

PAYMENT VALUE MIX



Credit cards contribute 78.46% of delivered-order payment value and average 3.5 installments. Authorization, installment cost, and credit-card conversion are material commercial dependencies.

CUSTOMER REVENUE MIX



RECOMMENDED TEST

Build a second-purchase lifecycle test after delivery: category-aware recommendations, timed incentive, and service recovery for delayed first orders. Measure incremental second-order conversion and contribution margin.

SQL Architecture and Delivery

The project demonstrates controlled analysis, not isolated syntax exercises

PATTERNS USED

CTES Reusable category, review, seller-order, customer, and basket layers.

WINDOW FUNCTIONS Running category revenue and payment/revenue shares.

FILTERED AGGREGATES Late-delivery rates with explicit eligible denominators.

GRAIN BRIDGES Order-level review and seller-order metrics before aggregation.

SELF-JOINS Distinct order-category pairs without symmetric duplicates.

IDENTITY DESIGN customer_unique_id for person-level repeat behavior.

DELIVERY CONTRACT

- PostgreSQL DDL, indexes, CSV load commands, and validation SQL.
- Ten numbered business queries with matching result extracts.
- DuckDB runner for server-free reproduction and automated row checks.
- Findings document with exact metrics, implications, and limitations.
- ERD and modeling notes explaining fact paths and cardinality risks.
- Executive PDF that communicates the analysis without requiring database access.

PORTFOLIO SIGNAL

A reviewer can inspect the SQL, rerun the project, reconcile every headline number to a CSV, and understand why each join is safe. That combination is the core analytics-engineering value of the case study.